Review Exam 1 — Solutions

EE 999 — Introduction to Control Systems

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Answers to ReviewExam1.pdf

Problem 1.1(a) — Signals and parameters

Model the DC motor drive of figure 1. The input is the applied voltage $V_a(t)$; the output is the shaft position $\theta(t)$.

Signal	Units	Description
$V_a(t)$	[V]	Applied armature voltage (input)
$i_a(t)$	[A]	Armature current
$v_L(t)$	[V]	Voltage across L_a
$v_R(t)$	[V]	Voltage across R_a
$E_b(t)$	[V]	Back EMF of the motor
$T_m(t)$	[N-m]	Torque produced by the motor
$M_k(t)$	[N-m]	Torque of the rotational spring
$\theta(t)$	[rad]	Shaft angle (output)
$\dot{\theta}(t) = \omega(t)$	[rad/s]	Shaft angular velocity

Table 1: Signals of the DC motor drive.

Parameter	Units	Description
L_a	[H = V-s/A]	Armature inductance
R_a	[Ω = V/A]	Armature resistance
k_t	[N-m/A]	Motor torque constant
k_e	[V-s/rad]	Motor back-EMF constant
J	[kg-m^2]	Rotor and load inertia
k	[N-m/rad]	Rotational spring stiffness

Table 2: Parameters of the DC motor drive.

Problem 1.1(b) — Constituent equations

$$v_L(t) = L_a \frac{di_a(t)}{dt} \tag{1}$$
$$v_R(t) = R_a\, i_a(t) \tag{2}$$
$$E_b(t) = k_e\, \dot{\theta}(t) \tag{3}$$
$$T_m(t) = k_t\, i_a(t) \tag{4}$$
$$M_k(t) = k\, \theta(t) \tag{5}$$
$$J\ddot{\theta}(t) = M_{net}(t) \tag{6}$$

Problem 1.2(a) — Free body diagram

The only body is the inertia J. The motor torque T_m acts in the positive θ direction, and the spring, whose far end is grounded, resists the rotation with torque $M_k = k\theta$.

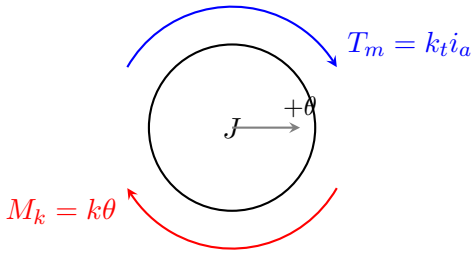


Figure 1: Free body diagram of the inertia J.

Problem 1.2(b) — Continuity equations

Kirchhoff's voltage law around the armature loop:

$$V_a(t) = v_L(t) + v_R(t) + E_b(t) \tag{7}$$

Sum of torques on the inertia:

$$J\ddot{\theta}(t) = T_m(t) - M_k(t) \tag{8}$$

Problem 1.3 — Governing equations in standard form

Substituting the constituent equations into the continuity equations:

$$L_a \frac{di_a(t)}{dt} + R_a\, i_a(t) = V_a(t) - k_e\, \dot{\theta}(t) \tag{9}$$
$$J\ddot{\theta}(t) + k\, \theta(t) = k_t\, i_a(t) \tag{10}$$

Eliminating $i_a(t)$ (done in the frequency domain in part 4) gives a single third-order equation with the output on the left and the input on the right:

$$L_a J\, \dddot{\theta}(t) + R_a J\, \ddot{\theta}(t) + (L_a k + k_t k_e)\, \dot{\theta}(t) + R_a k\, \theta(t) = k_t\, V_a(t) \tag{11}$$

Problem 1.4 — Transfer function $G_{uy}(s) = \theta(s)/V_a(s)$

With zero initial conditions:

$$(L_a s + R_a)\, I_a(s) = V_a(s) - k_e s\, \theta(s) \tag{12}$$

$$(Js^2 + k)\, \theta(s) = k_t\, I_a(s) \quad \Rightarrow \quad I_a(s) = \frac{Js^2 + k}{k_t}\, \theta(s) \tag{13}$$

Substituting $I_a(s)$ into the electrical equation:

$$(L_a s + R_a)\, \frac{Js^2 + k}{k_t}\, \theta(s) + k_e s\, \theta(s) = V_a(s) \tag{14}$$

$$\Big[(L_a s + R_a)(Js^2 + k) + k_t k_e s\Big]\, \theta(s) = k_t\, V_a(s) \tag{15}$$

In rational polynomial form:

$$\boxed{G_{uy}(s) = \frac{\theta(s)}{V_a(s)} = \frac{k_t}{L_a J\, s^3 + R_a J\, s^2 + (L_a k + k_t k_e)\, s + R_a k}} \tag{16}$$

Problem 1.5 — Units balance

Using $\mathrm{N \cdot m} = \mathrm{kg \cdot m^2/s^2}$, $[s] = 1/\mathrm{s}$ and radians dimensionless, every term of the denominator has the same units:

$$L_a J\, s^3 : \left[\tfrac{\mathrm{V\,s}}{\mathrm{A}}\right]\left[\mathrm{kg\,m^2}\right]\left[\tfrac{1}{\mathrm{s^3}}\right] = \tfrac{\mathrm{V}}{\mathrm{A}}\,\mathrm{N\,m}$$

$$R_a J\, s^2 : \left[\tfrac{\mathrm{V}}{\mathrm{A}}\right]\left[\mathrm{kg\,m^2}\right]\left[\tfrac{1}{\mathrm{s^2}}\right] = \tfrac{\mathrm{V}}{\mathrm{A}}\,\mathrm{N\,m}$$

$$L_a k\, s : \left[\tfrac{\mathrm{V\,s}}{\mathrm{A}}\right]\left[\tfrac{\mathrm{N\,m}}{\mathrm{rad}}\right]\left[\tfrac{1}{\mathrm{s}}\right] = \tfrac{\mathrm{V}}{\mathrm{A}}\,\mathrm{N\,m}$$

$$k_t k_e\, s : \left[\tfrac{\mathrm{N\,m}}{\mathrm{A}}\right]\left[\tfrac{\mathrm{V\,s}}{\mathrm{rad}}\right]\left[\tfrac{1}{\mathrm{s}}\right] = \tfrac{\mathrm{V}}{\mathrm{A}}\,\mathrm{N\,m}$$

$$R_a k : \left[\tfrac{\mathrm{V}}{\mathrm{A}}\right]\left[\tfrac{\mathrm{N\,m}}{\mathrm{rad}}\right] = \tfrac{\mathrm{V}}{\mathrm{A}}\,\mathrm{N\,m}$$

The numerator k_t has units $\mathrm{N\,m/A}$, so

$$[G_{uy}] = \frac{\mathrm{N\,m/A}}{\mathrm{V\,N\,m/A}} = \frac{1}{\mathrm{V}} = \frac{\mathrm{rad}}{\mathrm{V}} = \frac{[\theta]}{[V_a]}$$

which are the units of output over input, so the transfer function balances.

Problem 2.1 — Governing differential equation

$$G(s) = \frac{Y(s)}{U(s)} = \frac{3s+4}{2s^2+14s+20}$$

Cross-multiplying, $(2s^2+14s+20)\,Y(s) = (3s+4)\,U(s)$, and each power of s is a derivative:

$$\boxed{2\ddot{y}(t) + 14\dot{y}(t) + 20\,y(t) = 3\dot{u}(t) + 4\,u(t)} \tag{17}$$

Problem 2.2 — Impulse response

With $u(t) = \delta(t)$, $U(s) = 1$ and $Y(s) = G(s)$. In monic form with K_{rlg}:

$$Y(s) = \frac{3}{2}\,\frac{s+\frac{4}{3}}{s^2+7s+10} = \frac{3}{2}\,\frac{s+\frac{4}{3}}{(s+2)(s+5)} = \frac{C_1}{s+2} + \frac{C_2}{s+5} \tag{18}$$

By cover-up:

$$C_1 = \left[\frac{3}{2}\,\frac{s+\frac{4}{3}}{s+5}\right]_{s=-2} = \frac{3}{2}\cdot\frac{-\frac{2}{3}}{3} = -\frac{1}{3} \tag{19}$$

$$C_2 = \left[\frac{3}{2}\,\frac{s+\frac{4}{3}}{s+2}\right]_{s=-5} = \frac{3}{2}\cdot\frac{-\frac{11}{3}}{-3} = \frac{11}{6} \tag{20}$$

(Check: $C_1 + C_2 = \frac{3}{2}$, the leading coefficient of the monic numerator, as it must be for a strictly proper second-order system.)

$$\boxed{y(t) = \left(\frac{11}{6}\,e^{-5t} - \frac{1}{3}\,e^{-2t}\right)1^+(t)} \tag{21}$$

Problem 3.1(a) — Reference response $T_{ry}(s)$

$$G_c(s) = k_p, \quad G_p(s) = \frac{2s+3}{s(2s+1)}, \quad H_y(s) = \frac{1}{0.1s+1}$$

Signals around the loop, with $d = 0$:

$$E(s) = R(s) - H_y(s)\,Y(s) \tag{22}$$
$$Y(s) = G_p(s)\,G_c(s)\,E(s) \tag{23}$$

$$Y(s) = G_pG_c\big(R(s) - H_yY(s)\big) \tag{24}$$
$$\big(1 + G_cG_pH_y\big)\,Y(s) = G_cG_p\,R(s) \tag{25}$$
$$T_{ry}(s) = \frac{G_cG_p}{1 + G_cG_pH_y} \tag{26}$$

Substituting and clearing fractions (multiply top and bottom by $s(2s+1)(0.1s+1)$):

$$T_{ry}(s) = \frac{k_p\,(2s+3)(0.1s+1)}{s(2s+1)(0.1s+1) + k_p(2s+3)} \tag{27}$$
$$= \frac{k_p\,(0.2s^2 + 2.3s + 3)}{0.2s^3 + 2.1s^2 + (1+2k_p)\,s + 3k_p} \tag{28}$$

or, with monic polynomials,

$$\boxed{T_{ry}(s) = k_p\,\frac{s^2 + 11.5s + 15}{s^3 + 10.5s^2 + (5+10k_p)\,s + 15k_p}} \tag{29}$$

Problem 3.1(b) — Disturbance to error $T_{de}(s)$

With $r = 0$:

$$E(s) = -H_y(s)\,Y(s) \tag{30}$$
$$Y(s) = G_p(s)\big(D(s) + G_c(s)\,E(s)\big) \tag{31}$$

$$E(s) = -H_yG_p\,D(s) - H_yG_pG_c\,E(s) \tag{32}$$
$$\big(1 + G_cG_pH_y\big)\,E(s) = -G_pH_y\,D(s) \tag{33}$$
$$T_{de}(s) = \frac{-G_pH_y}{1 + G_cG_pH_y} \tag{34}$$

Multiplying top and bottom by $s(2s+1)(0.1s+1)$:

$$T_{de}(s) = \frac{-(2s+3)}{0.2s^3 + 2.1s^2 + (1+2k_p)\,s + 3k_p} \tag{35}$$

$$\boxed{T_{de}(s) = -10\,\frac{s + 1.5}{s^3 + 10.5s^2 + (5+10k_p)\,s + 15k_p}} \tag{36}$$

Both transfer functions share the same closed-loop denominator, $1 + G_cG_pH_y$, as expected.

Problem 4 — Laplace transform vs. transfer function

	Statement	Answer
1	It depends on system parameters.	T.F.
2	It is determined from a time domain signal.	L.T.
3	It has an inverse.	L.T.
4	It has units.	Both
5	The time-domain function can be found by partial fraction expansion.	L.T.
6	It is a rational polynomial.	Both

Table 3: Answers to Problem 4.

1. A transfer function is built from the masses, springs, resistances and so on of the system. A Laplace transform is built from a signal, which need not come from any system.

2. A Laplace transform is computed from a time-domain signal, $F(s) = \int_0^\infty f(t)e^{-st}dt$. A transfer function is derived from the differential equations of the system.

3. The inverse of a Laplace transform is the original time-domain signal. A transfer function describes a system, not a signal, so there is no time function it is "the transform of" (Part 3b, section 10).

4. A Laplace transform of $x(t)$ in meters has units of meter-seconds. A transfer function has the units of output over input, for example rad/V in Problem 1.

5. Partial fraction expansion is the inverse Laplace transform: it breaks $Y(s)$ into table entries to recover the signal $y(t)$. (A transfer function is only expanded this way once it has been multiplied by an input to form $Y(s) = G(s)U(s)$, as with the impulse in Problem 2.)

6. Both are ratios of polynomials in s: the transfer function comes from the coefficients of the differential equation, and every elementary signal in the Laplace table has a rational transform.

Problem EC — Dynamic equivalent

Using the Firestone analogy (velocity ↔ voltage, force ↔ current, mass ↔ capacitance, damper ↔ conductance $1/R$, spring ↔ inverse inductance $1/L$), each mass becomes a node whose voltage is its velocity, with a capacitor from that node to ground (the inertial reference).

1. Components:
   (a) Resistors: **2** — b_1 (to ground) and b_2 (between the masses).
   (b) Capacitors: **2** — M_1 and M_2.
   (c) Inductors: **1** — k_2.
2. **Two** non-ground nodes, one per mass: $v_1 = \dot{x}(t)$ and $v_2 = \dot{y}(t)$.
3. The applied force $f(t)$ becomes a **current source** $i_s(t) = f(t)$ driving node 1 from ground.
4. The schematic is shown in figure 2.
5. Component values:

Component	Value	Connected between
$i_s(t)$	$f(t)$	ground → node 1
C_1	M_1	node 1 – ground
R_1	$1/b_1$	node 1 – ground
L_2	$1/k_2$	node 1 – node 2
R_2	$1/b_2$	node 1 – node 2
C_2	M_2	node 2 – ground

Table 4: Component values of the equivalent circuit.

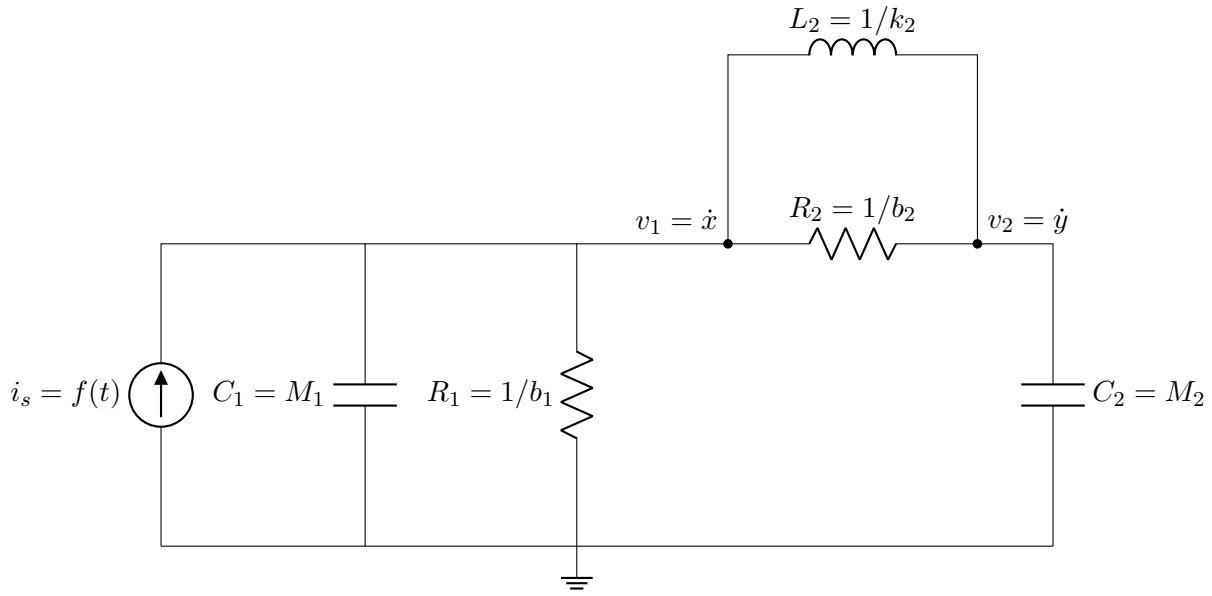


Figure 2: Firestone-analogous circuit of the mass with passive vibration damper.

Check by node equations. KCL at node 1 and node 2:

$$f = C_1\dot{v}_1 + \frac{v_1}{R_1} + \frac{1}{L_2}\int (v_1 - v_2)\,dt + \frac{v_1 - v_2}{R_2} \tag{37}$$

$$0 = C_2\dot{v}_2 + \frac{1}{L_2}\int (v_2 - v_1)\,dt + \frac{v_2 - v_1}{R_2} \tag{38}$$

which, with $v_1 = \dot{x}$, $v_2 = \dot{y}$ and the values above, are exactly the equations of motion of the two masses:

$$M_1\ddot{x} + b_1\dot{x} + k_2(x - y) + b_2(\dot{x} - \dot{y}) = f(t) \tag{39}$$

$$M_2\ddot{y} + k_2(y - x) + b_2(\dot{y} - \dot{x}) = 0 \tag{40}$$